



EXAONE Forecast

A Universal Zero-shot Time-Series Foundation Model

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We present **EXAONE Forecast**, a zero-shot time-series foundation model built on an encoder-only Transformer. A single model serves univariate, multivariate, and covariate-informed forecasting. Group attention placed only in the last layer exchanges cross-variate information at small cost, and a query-aware scalable softmax maintains sharp attention over long contexts. To pretrain the model at scale, we propose CKD-GP, a generator that expands composite Gaussian-process kernels and samples every term in closed form within $O(T \log T)$ time. We pretrain EXAONE Forecast on this synthetic corpus along with real, augmented, and coupled multivariate data. On GIFT-Eval, EXAONE Forecast achieves an aggregate MASE of 0.673 and CRPS of 0.460, ranking first in the zero-shot category on aggregate MASE and CRPS as well as on both rank-based scores. Additionally, the retrieval-augmented EXAONE Forecast-Agent takes first place in the agentic category in aggregate MASE and CRPS.

1 Introduction

Time-series forecasting supports decision making across domains such as demand forecasting and traffic forecasting in which future outcomes must be anticipated from historical observations. These applications have relied on dedicated models that are fitted to the history of their own domain and dataset [16, 17, 36]. This practice scales poorly, since a practitioner may face thousands of heterogeneous series and cannot retrain a model for each of them. Inspired by the success of large language models, time-series foundation models (TSFMs) instead pretrain a single model on large corpora that span many domains [3, 8, 31, 40]. Trained at scale in both data and parameters, such models deliver accurate zero-shot forecasts on unseen series without any retraining. Recent models such as Chronos-2 [4], TiRex-2 [28], Toto-2.0 [19], and TimesFM-2.5 [8] now match or surpass task-specific models on public benchmarks [2, 34].

Scaling such corpora with real data alone is challenging in the time-series domain. Publicly available collections remain limited in size and diversity, and many industrial series cannot be released [2, 3]. TSFM pipelines therefore rely heavily on simulated data, and Gaussian processes (GPs) are the dominant simulator [30]. KernelSynth [3] and CauKer [42] are prominent examples of GP-based generators, and recent models such as PatchTST-FM-r1 [37] and FlowState [14] rely on these two generators alone for their simulated data. Generating such corpora at scale is bottlenecked by the GP sampler itself. An exact draw of length T requires factorizing a $T \times T$ covariance matrix at $O(T^3)$ time and $O(T^2)$ memory. The factorization also becomes numerically unstable for smooth kernels and long contexts, so many draws fail outright even with jitter. As context lengths grow into the thousands and corpora into tens of millions of series, this step dominates data-generation cost and caps sequence length.

In this report, we present **EXAONE Forecast**, a zero-shot time-series forecasting model built on an encoder-only Transformer. EXAONE Forecast supports univariate, multivariate, and covariate-informed forecasting within a single architecture. It incorporates both past and future-known covariates by treating them as additional variates, and exchanges cross-variate information through group attention placed only in the final layer, keeping the additional computational cost small. In addition, it uses a query-aware scalable softmax to maintain sharp attention over long contexts. To train the model with GP-based simulation data at scale, we propose CKD-GP (**C**omposite **K**ernel **D**ecomposition for **GP** generation). CKD-GP expands each composite kernel into a sum of product terms and samples every term with a closed-form $O(T \log T)$ route. After analyzing the coverage and fidelity of the resulting synthetic corpora, we pre-

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train EXAONE Forecast on this generated data alongside real, augmented, and coupled multivariate data. We evaluate EXAONE Forecast on GIFT-Eval [2] and fev-bench [34]. Additionally, on GIFT-Eval, we build EXAONE Forecast-Agent, an agentic pipeline that combines multiple time-series foundation models through a learned router and a retrieval module.

Our contributions are summarized as follows:

- We present EXAONE Forecast, a single encoder-only TSFM that handles univariate, multivariate, and covariate-informed forecasting in a zero-shot manner. A last-layer group attention and a query-aware scalable softmax make cross-variate reasoning and long contexts affordable.
- We propose a framework for high-fidelity GP generation that decomposes composite kernels of KernelSynth and CauKer into terms amenable to fast sampling, achieving up to $2,645\times$ and $1,089\times$ speedups over dense sampling.
- We demonstrate strong zero-shot performance on public benchmarks. On GIFT-Eval, EXAONE Forecast ranks first in the zero-shot category on aggregate MASE and CRPS as well as on both rank-based scores, and EXAONE Forecast-Agent ranks first in the agentic category in aggregate MASE and CRPS. On fev-bench, EXAONE Forecast ranks among the top foundation models.

2 Related Work

Time-series foundation models. Recent TSFMs differ mainly in how they exchange information across variates and in the datasets used for pretraining. Chronos-2 [4] attaches covariates as extra variates and applies group attention over the variate axis, which enables in-context learning across related series. The Toto-2.0 family [7, 19] originates from observability data and alternates time-wise and variate-wise attention, while reporting that variate attention is needed only in a few layers. PatchTST-FM-r1 [37] revisits the PatchTST backbone [27] and reports several ablations on model width and depth, context length, and the number of quantile levels. TiRex-2 [28] extends the xLSTM-based TiRex [5] to multivariate inputs through synthetically coupled variates. FlowState [14] couples a state-space encoder with a functional basis decoder so that the same model adapts to any sampling rate. TimesFM-2.5 [8, 13] scales the decoder-only recipe with a longer context and a continuous quantile head. Recent work also incorporates textual information into forecasting: adaptive information routing [32] uses text to control how multivariate signals are combined within a forecasting model, while STRIDE [1] injects text-based reasoning to reduce prediction variance. EXAONE Forecast follows the temporal-wise Transformer encoder design [23] and combines patching [27], instance normalization, and quantile regression with the sparse group attention.

Agentic forecasting and model orchestration. Recent work has explored LLM agents as controllers that select, evaluate, or combine forecasting models rather than directly generating numerical predictions. TimeCopilot [12] provides a unified interface over statistical forecasters and multiple TSFMs, using an LLM agent to analyze time-series characteristics, select candidate models through cross-validation, and explain the resulting forecasts. MoiraiAgent [44] uses an LLM-based router to select among forecasting experts based on temporal characteristics, validation evidence, and available textual context. Conversational TSFMs [6] instead focus on interpretable ensemble coordination, training an LLM to reason about model weights and iteratively refine the forecasting strategy. Together, these studies frame model orchestration as an alternative to relying on a single TSFM. Within this direction, EXAONE Forecast-Agent combines input-dependent routing over a diverse model pool with retrieval of semantically related historical series.

Synthetic data generation. Synthetic pretraining data for time series traces back to GP priors. KernelSynth [3] samples kernels from a kernel bank and combines them with random $+$ and $\times$ operations before drawing one series from the composite kernel. CauKer [42] samples a causal graph, draws a GP series at each root node, and propagates these signals through the graph with non-linear link functions. All GP-based generators share the same bottleneck, namely a dense covariance factorization per draw. Random Fourier features [29], pathwise conditioning [39], and structured kernel interpolation [38] avoid this factorization by approximating the covariance. These approximate samplers, however, bound their error only through a feature or inducing-point budget. Circulant embedding [10, 41] instead draws exact stationary GP samples through the FFT. CKD-GP for EXAONE Forecast combines it with closed-form routes for periodic and linear kernels to cover the composite kernels used in practice.

TempoPFN [25] broadens the generator set with the ForecastPFN prior [11] and with stochastic differential equations such as the Ornstein–Uhlenbeck process. It further adds sawtooth, step, spike, and anomaly generators as well as audio-inspired generators. In our setting, however, these shape-based generators yielded little performance gain. To mitigate the limited diversity inherent in any single generator, additional augmentations are widely adopted. Common strategies include mixup across series and flipping along the time or value axes [3, 25]. For multivariate data, TiRex-

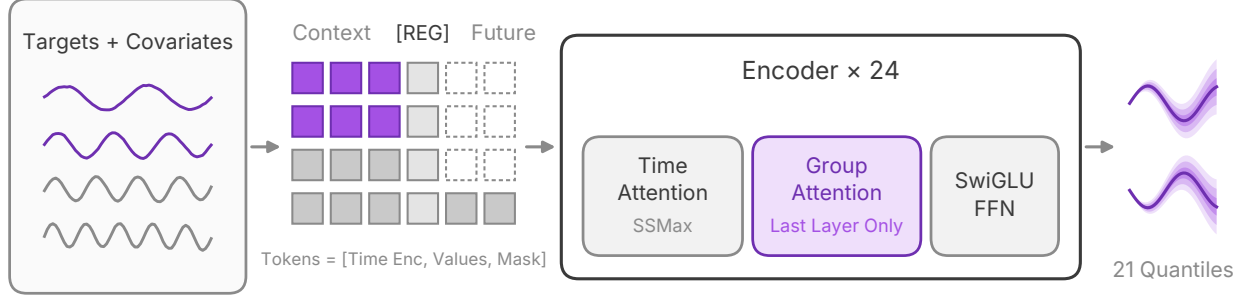


Figure 1: Overview of EXAONE Forecast. Targets and covariates form one group, and each variate becomes a token consisting of a row of context patches, a [REG] token, and future patches. Future tokens stay empty unless the covariate is known over the forecasting horizon. Every encoder block applies time attention with SSMx and a SwiGLU FFN, and group attention runs only in the last block. The quantile head emits 21 quantiles for every target.

2 [28] couples independently sampled univariate series with mechanisms such as linear mixing, cointegration, and structural causal models.

3 EXAONE Forecast

3.1 Problem Setup

We consider probabilistic forecasting of a target series $Y_{1:T} = [y_1, \dots, y_T]$ with $y_t \in \mathbb{R}^D$ over a horizon of H steps. The target is univariate when $D = 1$ and multivariate when $D > 1$. In addition, a series may come with M covariates $X_{1:T+H} = [x_1, \dots, x_{T+H}]$ with $x_t \in \mathbb{R}^M$. Past covariates are observed only up to time T , whereas future covariates such as calendar features or planned prices are known over the horizon as well (empty when unknown). The goal is to model the predictive distribution

$$P(Y_{T+1:T+H} | Y_{1:T}, X_{1:T+H}), \quad (1)$$

in a zero-shot manner. EXAONE Forecast represents this distribution by a set of quantile forecasts $\hat{y}_{T+h}^q$ at 21 levels $q \in \mathcal{Q} = \{0.01, 0.05, 0.10, \dots, 0.90, 0.95, 0.99\}$. Targets and covariates are treated uniformly as variates of one group. Every variate is embedded as its own token sequence, and cross-variate information is exchanged through group attention (Section 3.3). Future covariates fill the future patches of their variate, while the future patches of targets and past covariates are left empty. This design enables a single model to cover univariate, multivariate, and covariate-informed forecasting.

3.2 Input Pipeline

Scaling. Each variate d is normalized with the mean μ_d and the standard deviation σ_d of its observed context, followed by an arcsinh transform [4, 20]:

$$\tilde{v}_{t,d} = \sinh^{-1} \left(\frac{v_{t,d} - \mu_d}{\sigma_d} \right). \quad (2)$$

Here $v_{t,d}$ denotes the value of variate d at time t , whether target or covariate. The arcsinh behaves linearly near zero and logarithmically for large magnitudes, so extreme values are compressed without being clipped. Missing values are excluded from the statistics, and positions removed by the contiguous mask below are excluded as well. The same (μ_d, σ_d) normalizes the future covariates and the training targets of that variate. After prediction, the inverse transform $v_{t,d} = \sigma_d \sinh(\tilde{v}_{t,d}) + \mu_d$ maps the quantile forecasts back to the original scale.

Patching. The scaled context is split into non-overlapping patches of $P = 16$ time steps [27]. When the context length is not a multiple of P , the series is left-padded so that the most recent observations stay aligned with the patch boundary. The forecast horizon is likewise organized in output patches of P steps, and the model emits up to 32 output patches (512 steps) in a single forward pass. Longer horizons are handled by autoregressive unrolling, where a few quantile paths of the previous step are appended to the context and their predictions are merged by weighted quantile aggregation [4].

Table 1: Configuration of EXAONE Forecast.

Params	d_{model}	d_{embed}	Heads	Layers	Patch size	Quantiles	SwiGLU ratio	Max context
312M	1024	2048	16	24	16	21	2.667	8,192

Contiguous mask. During training, we apply contiguous patch masking (CPM) to the context. Blocks of $7 \times P = 112$ consecutive observations are dropped at random patch-aligned positions until about 2% of the context is masked. The masked positions are marked as missing. CPM exposes the model to long gaps that occur in real data, such as sensor outages, and discourages it from relying only on the most recent patch.

Input embedding. Each patch token is the concatenation of three P -dimensional vectors: a time encoding, the scaled values, and the observation mask. The time encoding assigns negative indices $[-L, \dots, -1]$ to context positions and $[0, \dots, H - 1]$ to future positions, both divided by the maximum context length. The $3P$ -dimensional token is projected to the model dimension by a residual MLP with a linear skip path and an RMSNorm [45] on the output. A learnable [REG] token separates context patches from future patches and acts as an aggregation point. The final token sequence of a variate is [context patches, [REG], future patches]. Future patches of targets contain zero values with an all-zero mask, so the encoder learns to fill them from the context and from the other variates.

3.3 Model Architecture

Model overview. EXAONE Forecast is an encoder-only Transformer that follows the temporal-wise design [23]. Every block applies pre-norm RMSNorm, a time attention with rotary position embeddings [35], and a SwiGLU feed-forward network with an expansion ratio of 2.667 [33]. The last block additionally applies a group attention over the variate axis. Table 1 lists the model configuration, which has 24 layers and 312M parameters. The encoder output at each future patch is passed to the quantile head, so that all output patches are predicted in parallel.

Time attention. Time attention captures temporal dependencies among the tokens of a single variate along the time axis. With a context of 8,192 steps and 32 output patches, one variate spans up to 545 tokens, and the attention distribution flattens as the number of keys n grows. This attention fading is a known failure mode of the softmax at long lengths [26]. Scalable softmax (SSMax) counteracts it by multiplying the logits with $\text{slog } n$, which keeps the sharpest attention weight bounded away from zero. We adopt a query-aware variant that scales each query before the dot product:

$$\tilde{q} = q \cdot \alpha_h(\log n) \cdot (1 + \tanh(g(q))), \quad (3)$$

where α_h is a small MLP that maps $\log n$ to a per-head scale and g is a query MLP that produces a scalar modulation. α_h is initialized to output one and g is zero-initialized, so training starts from the standard softmax. Since the scale is applied to q , the attention can still use Flash Attention kernels without any change.

Group attention. Group attention exchanges information across variates that belong to the same task, such as the channels of a multivariate series or a target and its covariates. The token tensor is transposed so that attention runs over the batch axis at each time position. A group-time mask $M_{q,b,t} = \mathbb{1}[g_q = g_b] \cdot m_{b,t}$ restricts each query to the variates of its own group g at observed positions m . Chronos-2 interleaves group attention with time attention in every block [4]. Toto-2.0 reports that cross-variate attention contributes little per layer and is best used sparsely [19], and we follow this observation by applying group attention only in the last block. This choice removes 23 of 24 variate-axis attentions and reduces the cost of long multivariate inputs accordingly. To keep the model robust to unseen group sizes, each group training task is subsampled with probability 0.5. When subsampling occurs, the number of retained variates is drawn uniformly from $\{1, \dots, n - 1\}$ for a task with n variates, so the same series appears as either a univariate task or a smaller group across different training steps.

Quantile head. The quantile head is a residual MLP that maps each future-patch embedding to $P \times |\mathcal{Q}|$ values. Its output is rearranged into a $|\mathcal{Q}| \times H$ quantile matrix and mapped back to the original scale. Quantiles that were not trained on are obtained by linear interpolation between neighboring levels.

Table 2: Pretraining corpus and sampling probabilities. The three real sources share the combined probability of 0.20, the remainder after the other sources. Each variate of a multivariate series counts as one series, and # Obs. sums the observations over these variates.

Source	Content	# Series	# Obs.	Prob.
Chronos	24 datasets of Chronos-1 [3]	0.87M	7.28B	0.20
GIFT-Eval Pretrain	53 datasets of the pretraining split [2]	2.38M	15.68B	
QuitoBench Corpus	CloudOps traffic of Alipay at 1h and 10m [43]	0.17M	1.57B	
Synthetic	KernelSynth, CauKer, ForecastPFN, OU process	78.92M	1,242.63B	0.20
Augmented	mixup and augmentation over all pools	53.05M	460.14B	0.50
Cross-Variate	coupled multivariate series, multivariate CauKer	41.74M	284.47B	0.10

3.4 Training

Training objectives. EXAONE Forecast is trained with the quantile (pinball) loss [21] over all levels and horizon steps:

$$\mathcal{L} = \frac{2}{|\mathcal{Q}|H} \sum_{q \in \mathcal{Q}} \sum_{h=1}^H m_h |\tilde{y}_{T+h} - \hat{y}_{T+h}^q| \cdot |\mathbb{I}[\tilde{y}_{T+h} \leq \hat{y}_{T+h}^q] - q|, \quad (4)$$

where $\tilde{y}$ is the target scaled with the context statistics and m_h masks missing targets. Future covariates are known in advance, so their future patches are excluded from the loss as well. The number of output patches is sampled uniformly from $\{1, \dots, 32\}$ for every batch, which trains the model on horizons from 16 to 512 steps.

Training details. We use the schedule-free AdamW optimizer [9] with a learning rate of 1.5×10^{-4} , $\beta = (0.9, 0.95)$, and 4,000 warm-up steps. A weight decay of 0.1 is applied to all weights except biases, normalization scales, and embeddings. Training runs on eight NVIDIA RTX Pro 6000 GPUs in bf16 mixed precision with a batch size of 48 per GPU. Batches are assembled under a token budget of $48 \times 8,192$ steps, so a task with many variates fills a batch alone with a proportionally shorter context. Datasets are sampled with fixed mixing probabilities (Section 3.5). Context length follows a three-stage curriculum: 2,048 steps for the first 20k updates, 4,096 steps until 40k updates, and 8,192 steps afterwards. The short contexts of the early stages enable a higher number of updates per unit of compute, while the final stage adapts the attention scales of Equation (3) to the full length. The full training run takes about 402 hours on this hardware.

3.5 Data Composition

Table 2 summarizes the pretraining corpus of EXAONE Forecast. It combines real data, synthetic data, augmented data, and synthetically coupled multivariate data. For zero-shot evaluation, the GIFT-Eval TrainTest source, comprising 55 train splits of the benchmark datasets, and its derived mixup data are excluded from the real and augmented pools.

Real data. Real data comes from three public collections: the Chronos corpus [3], the pretraining split of GIFT-Eval [2], and the QuitoBench corpus [43]. They span domains such as energy, transport, climate, and web operations over frequencies from minutes to years.

Synthetic data: structural priors. The synthetic pool combines non-GP and GP-based generators. Among the non-GP generators, the ForecastPFN prior [11] multiplies trend, seasonality, and noise components. The Ornstein–Uhlenbeck process adds mean-reverting stochastic dynamics driven by Brownian noise [25].

Synthetic data: Gaussian process family. KernelSynth [3] and CauKer [42] are the two GP-based generators that dominate recent TSFM corpora. Both draw kernels from a shared kernel bank. KernelSynth composes 1–5 kernels and draws one series per composite kernel, whereas CauKer composes 1–7 kernels per root node of a random DAG and propagates the draws through random activations. In both cases the exact GP draw dominates the cost of the whole corpus. We generate them with CKD-GP, which we describe next.

High-fidelity GP generation with CKD-GP. A GP on a regular grid induces $\mathbf{f} \sim \mathcal{N}(\mathbf{0}, \mathbf{K})$ with $K_{ij} = k(t_i, t_j)$, and the standard sampler factorizes $\mathbf{K}$ to return $\mathbf{L}\epsilon$. CKD-GP avoids this factorization through two closure properties of GPs. A GP with kernel $k_1 + k_2$ is the sum of independent GPs with k_1 and k_2 , and multiplying a draw from k by a deterministic $h(t)$ yields the kernel $h(t)h(t')k(t, t')$. CKD-GP therefore applies the distributive law to expand each

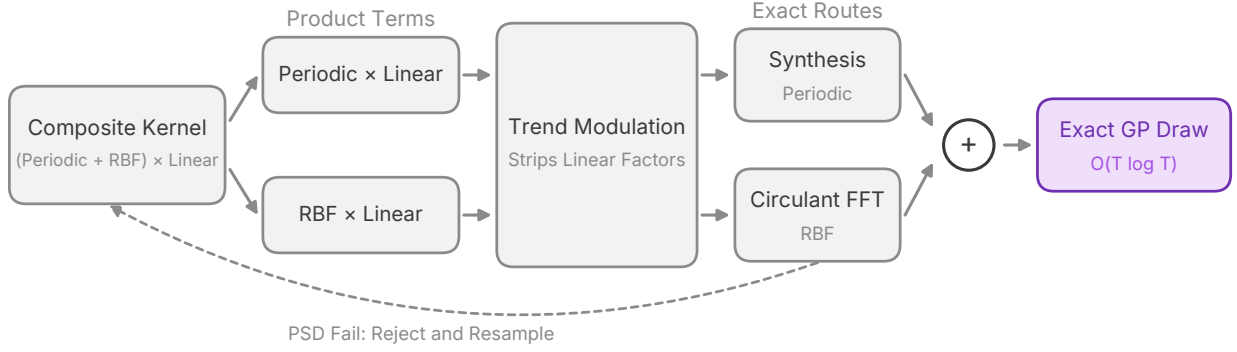


Figure 2: Overview of CKD-GP. A composite kernel expands into pure product terms. Trend modulation strips the linear factors from each term. The stationary remainders then route to harmonic synthesis (periodic) or the circulant FFT (decaying), and the draws sum into one faithful GP series in $O(T \log T)$ time. Draws that fail the PSD tolerance are rejected, and the kernel is resampled.

Table 3: GP sampling cost per accepted draw in milliseconds, averaged over 100 draws per length T . For CauKer, a draw denotes one accepted root-node sample. Cholesky is unsuitable as a baseline because composite kernels are often only positive semidefinite.

Method	Time	KernelSynth				CauKer			
		1024	2048	4096	8192	1024	2048	4096	8192
Dense eigh (CPU)	$O(T^3)$	238.3	1482.1	10787.7	82959.1	262.2	1650.0	11434.6	84907.0
Dense eigh (GPU)	$O(T^3)$	97.4	406.4	1924.0	10074.4	114.4	471.6	2251.7	11471.2
CKD-GP (CPU)	$O(T \log T)$	3.6	5.8	10.8	31.4	11.5	19.2	34.9	78.0
Speedup vs. CPU		66×	256×	999×	2645×	23×	86×	328×	1089×
Speedup vs. GPU		27×	70×	178×	321×	10×	25×	65×	147×

composite kernel into a sum of pure product terms. Figure 2 overviews the framework. Each term is then sampled by one of three closed-form routes.

1. *Non-stationary terms* carry linear (DotProduct) or constant factors. A lone linear or constant kernel is sampled in closed form, e.g., $f(t) = b + wt$ with Gaussian b and w . Otherwise trend modulation expands the d linear factors through their elementary symmetric polynomials, which reduces the term to at most $d+1$ stationary draws.
2. *Decaying stationary terms* that contain an RBF, rational-quadratic, or white-noise factor are sampled by circulant embedding [10, 41]. The Toeplitz covariance is embedded in a circulant matrix that the FFT diagonalizes, and complex Gaussian noise is transformed in $O(T \log T)$ time and $O(T)$ memory. The embedding uses analytically evaluated kernel values beyond the grid, and its size grows through factors $\{1, 2, 4, 8\}$ until the negative spectral mass falls below 10^{-3} .
3. *Purely periodic terms* do not yield a valid PSD embedding under our circulant construction, so they are synthesized from the exact Fourier series $k(\tau) = e^{-z} I_0(z) + 2 \sum_{j \geq 1} e^{-z} I_j(z) \cos(2\pi j \tau / p)$ with $z = \ell^{-2}$, truncated at 512 harmonics.

Draws whose embedding still fails at the largest size are rejected and the kernel is resampled, so a dense factorization is never performed. Series that are flat, divergent, or non-finite are rejected in the same way.

Table 3 compares the cost of CKD-GP with dense eigendecomposition. At $T = 8192$, CKD-GP delivers an accepted KernelSynth series in 31.4 ms against 83.0 s for the dense CPU route, a 2,645× speedup. CauKer is 2.5–3.3× slower than KernelSynth at equal length because it composes up to seven kernels and consequently produces more expanded terms. We analyze the generated corpora along two axes.

In terms of *coverage* (Table 4), 10.4% of KernelSynth draws and 14.8% of CauKer draws fail the PSD tolerance and are resampled, and enlargement beyond the minimal embedding is needed for about 42–44% of successful circulant calls. Nearly all failures arise from products of a periodic factor with a rational-quadratic or RBF factor. No draw

Table 4: Coverage statistics of the CKD-GP corpora (%). PSD fail is per draw, Enlarged per circulant call, and the last three per series. Brackets indicate the distribution of successful calls over the growth factors {1, 2, 4, 8}.

Corpus	PSD fail	Enlarged	NaN/Inf	Divergent	Flat
KernelSynth	10.43	43.60 [56.4/0.02/31.3/12.3]	0.00	0.00	0.63
CauKer	14.81	41.20 [58.8/0.00/29.1/12.2]	0.00	0.00	0.39

produced NaN or divergent output, and only 0.63% and 0.39% of series were flat. In terms of *fidelity*, the realized covariance of an accepted draw deviates from the analytic kernel by at most 9.9×10^{-4} on the circulant route and 3.1×10^{-2} on the harmonic route. The harmonic bound is the most permissive guarantee in the framework, since the 512-harmonic cap trades covariance fidelity for a $3.2\text{--}3.9\times$ reduction in tail (p_{99}) sampling cost. It is exact to 1.5×10^{-15} on the analytic and trend-modulation routes, whereas the dense `eih` route it replaces is itself accurate only to about 10^{-13} . Rejection is not distribution-preserving: kernels with a rational-quadratic factor appear 21.6% (KernelSynth) and 24.8% (CauKer) less often in the accepted corpus than in the proposed draws, while the other families shift by less than 10%.

Synthetic multivariate coupling. Due to the scarcity of real multivariate data and cross-variate data, group attention models struggle to learn effectively. Following TiRex-2 [28], we draw univariate series from all pools and couple them with a sampled mechanism. Six mechanisms are used. Identity stacking presents independent variates as one group, and linear mixing applies a random matrix with a sampled singular spectrum. Functional coupling derives covariates as noisy transforms of a base series, and cointegration adds shared random-walk trends to stationary deviations. Finally, linear and non-linear structural causal models combine lagged parents over a random DAG with log-uniform lags. A second cross-variate source runs CauKer in multivariate mode and emits the nodes of the sampled graph as one group of 2–100 variates.

Augmented data. The augmented pool follows the TempoPFN recipe [25]. Series drawn from all pools are mixed with mixup and then transformed by a random subset of augmentations. Regime changes, shock-and-recovery events, and seasonal amplitude modulation are injected on top to widen the range of non-stationary behaviors.

4 Results

We evaluate EXAONE Forecast on two public benchmarks. GIFT-Eval [2] covers 97 configurations from 23 datasets across seven domains and ten frequencies. Each dataset is evaluated at short, medium, and long horizons where permitted by its sequence length. fev-bench [34] covers 100 tasks featuring multivariate targets and covariates.

Metrics. GIFT-Eval reports both the Mean Absolute Scaled Error (MASE) and the Continuous Ranked Probability Score (CRPS). MASE divides the absolute error of the median forecast by the error of a seasonal naive forecast on the same series. CRPS scores the full predictive distribution and is approximated by the mean weighted quantile loss over nine levels. Both metrics are normalized by Seasonal Naive per configuration and aggregated with a geometric mean. The leaderboard additionally ranks every model on each configuration and averages the ranks, which yields MASE rank and CRPS rank. fev-bench scores each task with the scaled quantile loss (SQL), a CRPS variant scaled by the error of a naive baseline. It summarizes SQL in two ways: a skill score defined as one minus the geometric-mean SQL ratio against Seasonal Naive, and the average pairwise win rate across models. Since the rank metrics and the win rate are relative to the leaderboard model pool, their values can change as the pool changes.

4.1 GIFT-Eval

EXAONE Forecast. EXAONE Forecast is listed in the GIFT-Eval *zero-shot* category, since the GIFT-Eval TrainTest source and its derived mixup data are excluded from pretraining. It achieves an aggregate MASE of 0.673 and CRPS of 0.460 (Figure 3). Among the zero-shot category models, it ranks first on both aggregate metrics as well as on both rank-based scores.

EXAONE Forecast-Agent. EXAONE Forecast-Agent combines eight complementary pretrained models with retrieved historical cases. The model pool consists of Chronos-2 [4], TimesFM-2.5 [8, 13], FlowState [14], Toto-2.0-2.5B and its GIFT-Eval-adapted fine-tune Toto-2.0-2.5B-FT [19], TiRex-2 [28], Timer-S1 [22], together with EXAONE Forecast. A learned meta-router forms an instance-specific ensemble of their quantile forecasts.

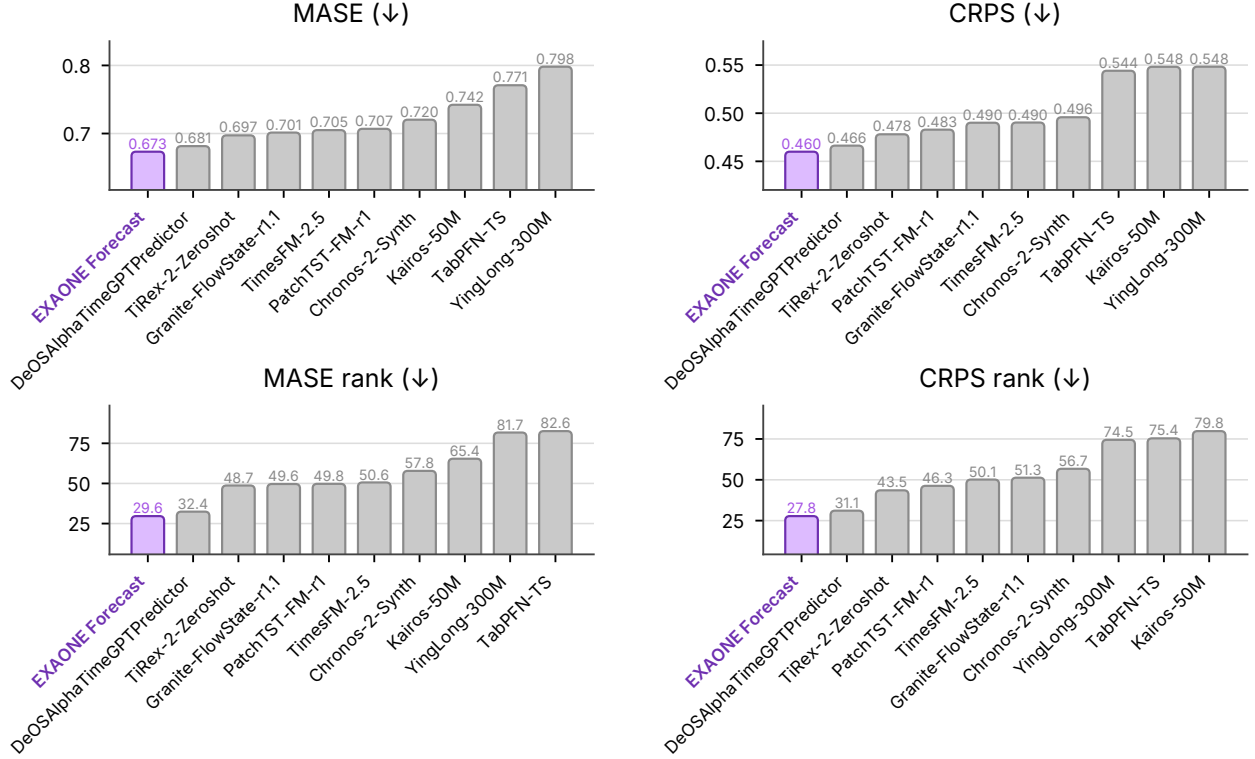


Figure 3: GIFT-Eval comparison with recent zero-shot foundation models.

Table 5: Ablation results for the EXAONE Forecast-Agent pipeline on GIFT-Eval. Metrics are aggregate scores relative to Seasonal Naive; lower is better.

Configuration	MASE	CRPS
Router	0.6504	0.4427
Router + Retrieval (EXAONE Forecast-Agent)	0.6099	0.4185

Router. Following the performance-based combination principle of FFORMA [24], the meta-router is trained on forecasting instances constructed from the available training splits. It assigns weights using four groups of input signals: statistics of the observed context, shape characteristics of the candidate forecasts, recent holdout performance of each model, and pairwise disagreement between model forecasts. The router is frozen at inference, and its weighted forecast is sorted along the quantile axis to ensure a valid ordering. The routed ensemble achieves a MASE of 0.6504 and CRPS of 0.4427 (Table 5) and serves as the base forecast for the retrieval stage.

Retrieval. Following retrieval-augmented forecasting [15, 18], the retrieval module targets recurring local patterns not represented precisely by the model pool. An LLM agent uses data descriptions and task metadata to select training series related by domain and measured quantity. Historical context–continuation pairs form a candidate bank, from which time-series similarity retrieves cases and converts their continuations into a retrieval forecast. The LLM only restricts this search space; it neither generates forecasts nor observes test targets.

A validation-based gate determines whether retrieval is suitable for each task. When enabled, the final forecast blends the routed and retrieval forecasts, with the retrieval contribution capped globally and adjusted for each instance according to match quality. For tasks in which retrieval fails to improve forecasts on the validation split, the gate returns the routed forecast unchanged. Router training, retrieval selection, and gate calibration use only the designated training data and their internal holdouts; test targets are not used at any stage.

Results. On GIFT-Eval, EXAONE Forecast-Agent achieves an aggregate MASE of 0.6099 and CRPS of 0.4185, ranking first in the agentic category on both metrics (Figure 4). Adding retrieval on top of the routed ensemble reduces MASE by 6.2%, from 0.6504 to 0.6099, and CRPS by 5.5%, from 0.4427 to 0.4185 (Table 5).

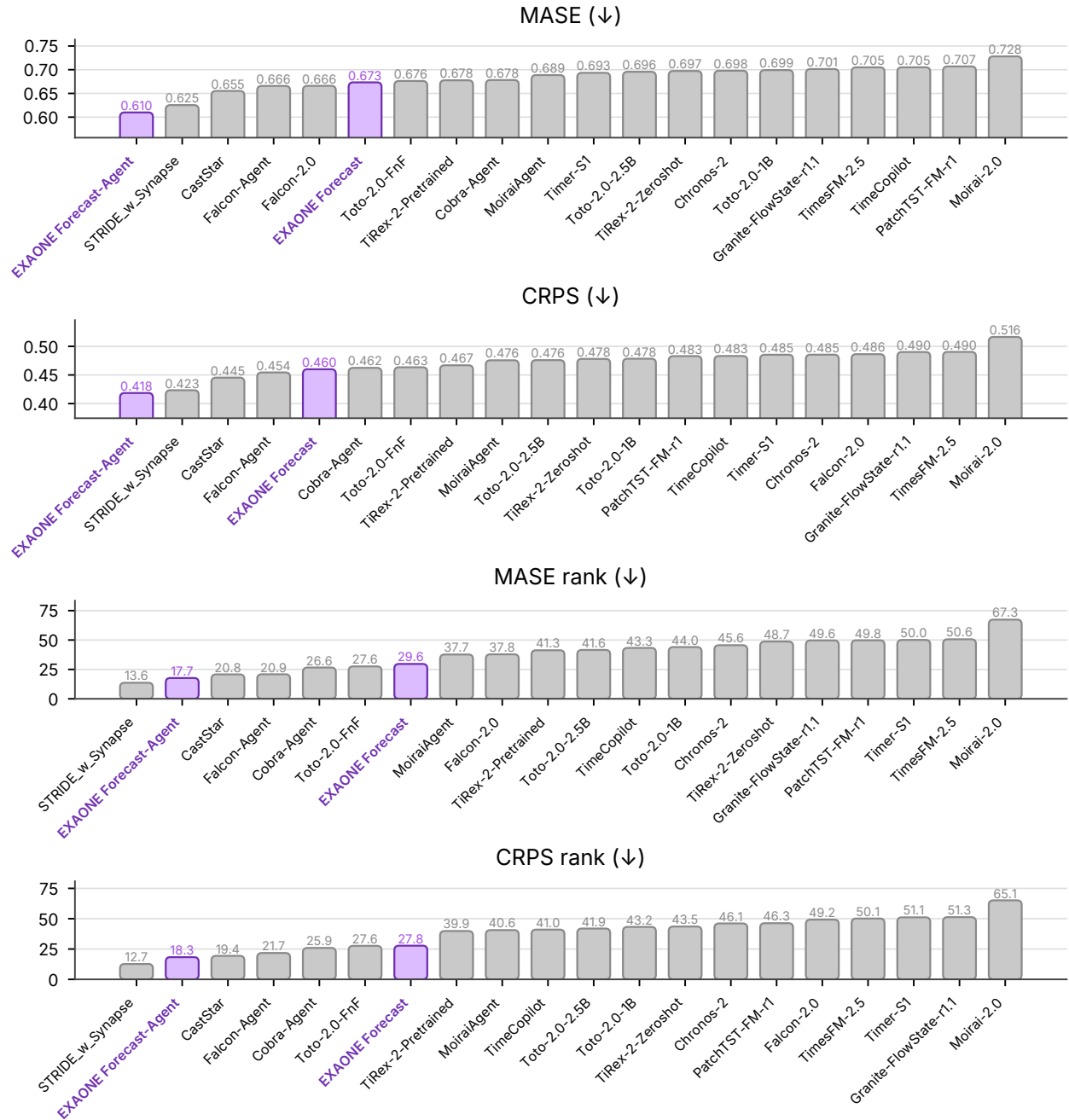


Figure 4: GIFT-Eval comparison of EXAONE Forecast and EXAONE Forecast-Agent with recent foundation models, including agentic forecasting methods.

4.2 fev-bench

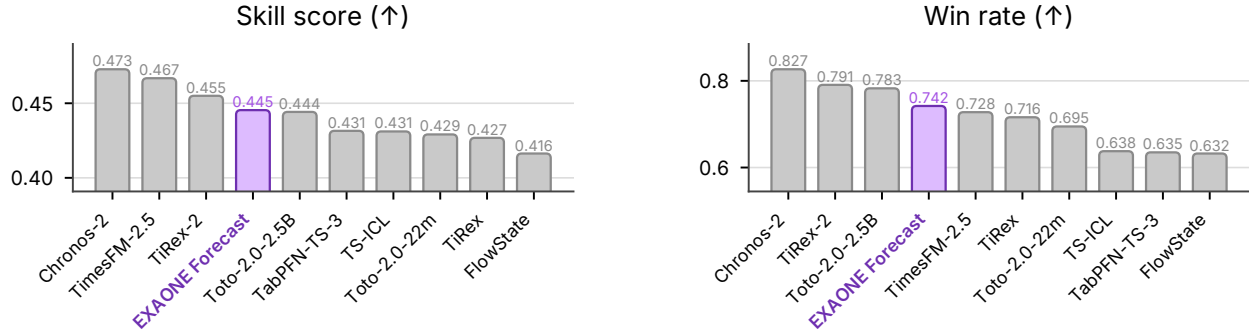


Figure 5: fev-bench comparison of EXAONE Forecast with recent foundation models.

fev-bench emphasizes the multivariate and covariate-informed abilities that GIFT-Eval covers only partially. Its 100 tasks include 46 with dynamic covariates and many with dozens of target variates. We report EXAONE Forecast (Figure 5) together with the published results of recent foundation models. EXAONE Forecast achieves a skill score of 0.445 and a win rate of 0.742. In terms of skill score, it ranks fourth among the models shown after Chronos-2, TimesFM-2.5, and TiRex-2. In terms of win rate, it also ranks fourth, behind Chronos-2, TiRex-2, and Toto-2.0-2.5B and ahead of TimesFM-2.5.

5 Conclusion

We presented EXAONE Forecast, an encoder-only foundation model for zero-shot forecasting, and CKD-GP, the generator behind its pretraining corpus. A single encoder covers univariate, multivariate, and covariate-informed forecasting, and the closed-form draws of CKD-GP scale high-fidelity pretraining data to tens of millions of long series. The benchmark results support our approach. The resulting model performs on par with recent TSFMs on GIFT-Eval and fev-bench. The agentic ensemble then demonstrates how retrieval can enhance routed forecasts, improving the overall MASE from 0.6504 to 0.6099 and the overall CRPS from 0.4427 to 0.4185. Several promising directions remain open for future work. Improving covariate-informed forecasting requires exposing the model to a more diverse set of synthetic covariates. Scaling the model is another direction and may call for recent language-model techniques such as mixture-of-experts.

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